Political Entrenchment and the Persistence of Economic Sanctions

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Abstract

We develop a dynamic model in which economic sanctions shift the balance of political power within the target country. Each period the target chooses how far to comply and the sender responds with sanctions. Sanctions raise political entrenchment, the de facto power of target-side constituencies aligned with continued confrontation, and both governments value this state. For a general class of economies, the model provides conditions under which myopia makes sanctions unravel: the target adjusts compliance just enough to induce sanction removal. When both governments are forward-looking, the target maintains non-compliance and sanctions persist indefinitely, even under complete information. The target's patience determines whether relief or non-compliance prevails, while a forward-looking sender raises the steady-state level of political entrenchment. Under stated conditions, a sufficiently long temporary sanction shock locks in persistent sanctions. The game alone does not pin down a welfare ranking. The core results are formally verified via machine-checked proofs in Lean 4.

Keywords: economic sanctions; persistence and hysteresis; dynamic political economy; feedback Stackelberg equilibrium; political entrenchment; path dependence.

JEL classification: C73; D72; F51; C61; D74.

1 Introduction

A striking feature of economic sanctions is how long some of them last. A sanctioning government restricts trade, financial, or diplomatic access to induce a target government to change its behavior, and such measures are typically framed as temporary instruments of coercion.[1] Yet many prominent sanction episodes, for example those against Cuba, Iran, and North Korea, have persisted for decades without fundamental change in the target government (Halcoussis et al., 2021, p. 162). This persistence is difficult to reconcile with a standard result in sanction theory: under complete information, targets concede at the threat stage, so actually imposed sanctions reflect either bargaining failure or incomplete information (Drezner, 2003). If persistence required such frictions, better information or more effective bargaining would end it. Why, then, can sanctions persist between optimizing governments that know each other's objectives, and how does each government's foresight shape the outcome?

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[1] Stated objectives can be wider: those of the U.S. measures on Russia are economic pressure, inhibition of the war effort, and accountability (U.S. Government Accountability Office, 2025, p. 14).

We focus on one domestic-political mechanism: sanctions can strengthen the domestic constituencies that favor their continuation. Government objectives incorporate the political influence of interest groups, and sanctions alter this influence by shifting a political state variable. This state, which we label political entrenchment, is the de facto power (in the sense of Acemoglu and Robinson, 2008, p. 268) of target-side constituencies whose interests are tied to continued confrontation. In their account, such power is not allocated by formal institutions but derives from a group's wealth, weapons, or ability to solve the collective action problem. It rises when sanctions exceed its inherited level and erodes when sanctions fall below it. The target incumbent values this state. We assume the sender's government also values it, modeling its domestic ruling coalition as being sustained by a visibly entrenched adversary (Section 2.2). Case studies of Iran sanctions illustrate both sides of this mechanism. Qualitative evidence documents that enterprises tied to the Revolutionary Guard accumulated wealth under comprehensive sanctions, while independent business owners lost revenue and influence—a shift in power toward constituencies aligned with confrontation; some within the ruling establishment favored continuing the sanctions (Bajoghli et al., 2024, pp. 51–52). In the sanctioning country, lifting sanctions can carry a domestic political cost that imposing them does not (Bajoghli et al., 2024, p. 118), a sender-side political dynamic captured by the model's sender payoff function. These qualitative accounts motivate our formal mechanism.

The two governments play a dynamic complete-information game. In each period, the target first chooses a level of compliance; the sender then observes this choice and sets the intensity of sanctions. Information is complete and monitoring is perfect, yet each government weighs both the current payoffs of its actions and the future political entrenchment that its actions help shape. We derive continuation values for both governments and verify that the sender's response is optimal following every feasible compliance choice—both on and off the equilibrium path—and that the target's choice is a best response at every state, as required for a stationary feedback Stackelberg equilibrium. The general results hold across a class of economies with general transition and payoff functions; closed-form solutions for a quadratic–linear specification within the class are reported in the Online Appendix.

Our analysis delivers three core sets of results. First, foresight sustains sanction persistence. When both governments are myopic—attaching no weight to future political entrenchment—the target accommodates: it adjusts compliance just enough to induce the sender to halt sanctions, and the entrenchment state erodes over time (Proposition 2). When both governments are forward-looking, the same economy supports an equilibrium where the target maintains non-compliance, sanctions exceed the myopic sender's level and never end, and the political state converges to a positive steady state (Theorem 3). The target's patience determines whether relief or non-compliance prevails; the sender's patience sets the compliance threshold for relief or, under non-compliance, raises both sanction intensity and the steady-state entrenchment level (Sections 3.2 and 5.1). Second, sanctions exhibit lock-in effects. Under stated equilibrium and transition conditions, once the political state enters a high basin—an interval of high entrenchment—equilibrium play keeps it there and sanctions never decline. A sufficiently long temporary sanction shock can move the state into this basin from any initial condition (Proposition 5). Third, the game alone does not pin down a welfare ranking (Section 4). These results hold at varying levels of generality. The benchmark comparison is derived from inequalities across model primitives on the class, holding all primitives fixed except the discount factors; lock-in holds at every equilibrium satisfying its conditions; and the welfare results apply across the class, with an explicit interest-group polity and the quadratic–linear family serving as illustrative examples.

Patient conflict and differences in patience are established themes; we study them in a target-first sanctions game in which the sender re-optimizes each period and its sanctions move a political state that both incumbents value. That a longer shadow of the future can sustain conflict when

today's outcome shapes tomorrow's position is established in the theory of conflict (Skaperdas and Syropoulos, 1996; Garfinkel and Skaperdas, 2000; Powell, 2012; Krainin and Wiseman, 2016), typically under one common discount factor. Differences in patience also have antecedents. Eaton and Engers (1992) let the two sides of a sanctions game discount differently, without an endogenous political state, and relate their toughness to patience and sanction costs; our order of moves adapts their game without its threat stage. McBride and Skaperdas (2009, Appendix B) let a long-lived party face one-period challengers. The earlier working paper Gong et al. (2025) studies a target-only dynamic program, with target patience and the partial-adjustment law as a leading example. A forward-looking sender with its own discount factor is what allows the two governments' forward-looking incentives to be separated.

Relative to the public-choice analysis of sanctions, the model makes the political feedback dynamic and strategic. Both governments' choices are derived from continuation values over a political state that the sanction itself moves. The public-choice starting point is Kaempfer and Lowenberg (1988), who explain sanctions through the political pressure and effectiveness of organized interest groups. Halcoussis et al. (2021) emphasize that sanctions may retain domestic political functions after their direct leverage has diminished. The model concerns how long sanctions last once imposed, and so complements empirical work on duration and termination (Bolks and Al-Sowayel, 2000; Attia et al., 2020; Attia and Grauvogel, 2023).

The model is closest to the work of Acemoglu and coauthors on persistent political power, and it differs from that work in who moves the power state and in what persists. In Acemoglu and Robinson (2008), an elite buys de facto power anew each period, and changes in it can partly or entirely offset changes in de jure power, the power that formal institutions allocate, so that economic institutions persist while political institutions change (pp. 268, 272, 287). In the framework of Acemoglu et al. (2005, Section 1.2), a group's resources shape its de facto power, which tends to reproduce the group's advantage unless shocks, including changes in the international environment, modify the balance. In Acemoglu et al. (2012, 2015), as in Bai and Lagunoff (2011), current decisions of those who hold power determine its future distribution. Domestic politics can also keep a costly conflict going: in weakly institutionalized polities, a civilian government whose rents a civil war leaves largely unaffected keeps its army too small to end the war, because a strong army opens the way to military coups (Acemoglu et al., 2010, p. 664). The model adds three things to this work. (i) The power state is a slow stock moved by a foreign government's policy: the sender's sanctions, a change in the target's international environment, raise or erode it, and what persists is that policy rather than a domestic institution. (ii) Two governments with separate discount factors act on the same state, so the roles of foresight come apart. In Acemoglu and Robinson (2008, p. 287), more forward-looking elites are more likely to dominate politics; here a more patient target sustains non-compliance whenever a less patient one does (Proposition 7), while a forward-looking sender raises the steady-state level of political entrenchment. (iii) A temporary sanction shock can carry the state into a high basin that equilibrium play does not leave (Proposition 5), and the game does not determine whether the resulting persistence is socially costly (Proposition 9), whereas in Acemoglu and Robinson (2008, p. 279) the elite's investment in de facto power is socially costly.

The model also relates to hysteresis and lock-in. Relative to hysteresis and lock-in models (Dixit, 1989; Krugman, 1991; Arthur, 1989; Hassler et al., 2003), persistence here is an equilibrium property of a game in which one government chooses the instrument that moves the state, against the other's response.[2]

Sanctions can persist here under complete information because a political state, which the sanc-

[2]The full positioning, including the earlier working paper and the analytical methods, is in Appendix O of the Online Appendix.

tions themselves move, carries each government's incentives from one date to the next. Information-based accounts explain much of the dynamics of conflict. In Acemoglu and Wolitzky (2014), conflict spirals end from inside: Bayesian players with limited memory come to treat a long spiral as uninformative about the opponent's type and experiment with cooperation. In the framework of Acemoglu and Wolitzky (2024), mistrust, misperception and misunderstanding drive conflict, and dates are linked by a belief about the opponent's type or a threat to switch equilibria. In complete-information models, valuing the future can also sustain conflict (Skaperdas and Syropoulos, 1996; Powell, 2012). The political-state mechanism is of that kind, with no belief to revise, and it is how the model answers the objection of Drezner (2003) stated at the outset. The same state can also end a conflict: after a temporary suspension of sanctions, a resumed sanction rule that stays at or below the state keeps de-escalation in place (Section 3.3). Appendix O.4 of the Online Appendix compares the two accounts in detail.

An Online Appendix and a formal supplement accompany the paper. The Online Appendix gives every proof and the precise statements of the results that the main text gives in words. The formal supplement relates the core results to a companion Lean 4 formalization, separately marks the analytic extensions that rely on external existence results, and gives a separately reviewed map from the formal statements to the paper's economic sentences.[3]

The paper makes three contributions. First, it separates what each government's foresight does to sanctions, holding every other primitive fixed. The comparison rests on the entrenchment condition, under which non-compliance is an investment: a sanction's pain today buys entrenchment worth at least as much, so a forward-looking target accepts the sanction rather than pay to end it, while a myopic one accommodates. The condition is sufficient for this, not necessary. A forward-looking sender sanctions a non-compliant target more than a myopic one, by a foresight premium with a closed-form lower bound (Section 3.2), and with a myopic sender a forward-looking target still maintains non-compliance and sanctions stay positive (Theorem 12). Several results reach beyond the constructed equilibria: a more patient target keeps a full-non-compliance equilibrium when the sender's response does not fall with the state and a condition on primitives holds (Proposition 7); in a linear–quadratic special case such an equilibrium exists exactly from a threshold discount factor that the sender's patience does not move (Proposition 6); under a stronger form of the entrenchment condition every equilibrium gives the target at least its pure-strategy minmax payoff, and on the class every equilibrium with a non-decreasing sanction rule has full non-compliance (Section 3.4); and no date-zero payment between the governments makes a temporary suspension of sanctions acceptable to both (Section 5.3). Second, it shows how sanctions entrench themselves. Lock-in holds at any equilibrium of any economy that meets explicit conditions; low sanctions can entrench themselves in the same way; when one equilibrium has both a high and a low basin, the initial state selects between them; and persistence need not mean hysteresis, since under further conditions every start converges to the same long-run state (Section 3.3). Third, it separates political persistence from social welfare. Two interest-group representations that reproduce the governments' payoffs, and so leave the equilibria unchanged, rank the forward-looking equilibrium and the benchmark in opposite social orders (Proposition 9). Under compliance neutrality some representation makes the play of every full-non-compliance equilibrium socially optimal (Proposition 11), while in an explicit interest-group polity in the tradition of Kaempfer and Lowenberg (1988) sanctions at one example economy stay at their maximum although ending them would maximize aggregate welfare. Evaluating welfare therefore requires group-level welfare primitives and an explicit aggregation in addition to the equilibrium.

[3]The Lean 4 development, the paper sources and the three documents are public at `https://github.com/qidaolake/political-entrenchment-sanctions-lean`.

The rest of the paper sets out the model, the results on the class and welfare. Section 2 defines the model and the example economies and solves the myopic benchmark. Section 3 compares forward-looking governments with it, bounds the sender's part of the difference, and presents lock-in and the role of the target's patience. Section 4 separates the governments' incentives from aggregate welfare. Section 5 solves the cases in which only one government looks forward, follows the path of the political state, interprets the results and states what empirical work would have to measure (Section 5.4), and Section 6 concludes.

2 Theoretical framework: sanctions and political entrenchment

2.1 Timing, actions, and the political state

Consider two governments, a target and a sender, that interact at dates $t = 0, 1, 2, \ldots$ and move in sequence at each date. Each date proceeds as follows.

(1) Both governments observe the political state $x_t \in [0, 1]$, defined below.

(2) The target chooses compliance $a_t \in [0, 1]$.

(3) The sender observes a_t and chooses a sanction intensity $s_t \in [0, 1]$.

(4) Each government receives its per-period payoff, which depends on (x_t, a_t, s_t) (Section 2.2).

(5) The state moves to x_{t+1} by the law of motion (1) below.

The order adapts the sanctions game of Eaton and Engers (1992) without its threat stage: the sender announces nothing in advance and chooses its sanction after observing compliance. The target therefore knows that its choice triggers the sender's response and, in the economies studied here, that less compliance makes the sender want to sanction more (Section 2.2). A period is one such review of compliance, and every result uses this order. Actions and the state are observed by both, and information is complete. The target and the sender discount their own payoffs at $\beta_T, \beta_S \in [0, 1)$. A government is *myopic* when its discount factor is zero and *forward-looking* when it is positive.

The state x_t is the de facto political power of the target's isolationist and anti-concession constituencies, whose position is tied to continued confrontation, and sanctions move it. A sanction above the inherited state raises it, and one below erodes it. Its law of motion is a reduced form for how that power evolves. The law is

$$x_{t+1} = F(x_t, s_t), \tag{1}$$

where $F\colon [0, 1]^2 \to [0, 1]$ satisfies the following conditions throughout.

(F1) F is continuous.

(F2) F is non-decreasing in the current state, so power persists.

(F3) F is non-decreasing in the sanction.

(F4) $F(x, x) = x$ for every x: a sanction equal to the current state reproduces it.

(F5) For every $x, s \in [0, 1]$, the state moves strictly toward the current sanction,

$$s < x \implies F(x, s) < x, \qquad x < s \implies x < F(x, s).$$

With (F2) and (F4), (F5) rules out overshooting,

$$s < x \implies s \leq F(x,s) < x, \qquad x < s \implies x < F(x,s) \leq s,$$

so zero sanctions strictly erode every positive state, $F(x,0) < x$ for $x \in (0,1]$, and full sanctions strictly raise every state below one, $x < F(x,1)$ for $x \in [0,1)$. Comparing state and sanction is a normalization within the model. Monotonicity of the law of motion in the state and in the sanction, (F2)–(F3), is the deterministic counterpart of the assumption, standard in stochastic games, that higher states and actions lead to higher states (Curtat, 1996, p. 185; Amir, 2002, p. 43). (F2) is the state dependence of political power of Acemoglu and Robinson (2008, p. 278) in a continuous state, and (F3) the rally effect by which a sanction can raise the political effort of the target regime's supporters (Kaempfer and Lowenberg, 1988, p. 791). An interval of states from a threshold up to 1 is a *high basin*, and one from 0 up to a threshold a *low basin*; the results ask when equilibrium play keeps the state in such an interval.

2.2 Objectives and the political stakes

We next turn to the governments' objectives: each government's per-period payoff depends on the political state, the target's compliance and the sanction.[4] Write $U_T, U_S \colon [0,1]^3 \to \mathbb{R}$ for the target's and the sender's per-period payoffs in (x,a,s), and V_T, V_S for their continuation values, which are bounded continuous functions of the state. The payoffs are continuous. They are government objectives: they weight the interest groups each government answers to, so they can differ from national welfare (Section 4). A *target policy* α gives compliance as a function of the state. *Full non-compliance* means $a = 0$, and the *full-non-compliance policy* $\alpha \equiv 0$ chooses it at every state; an equilibrium (Section 2.4) with this policy is a *full-non-compliance equilibrium*. *Full compliance* means $a = 1$, and the *full sanction* $s = 1$. *Relief* means $s = 0$, and *permanent relief* means $s = 0$ at every date. The target *accommodates* when its compliance brings the sanction to zero.

The results use no functional form for the transition or for either payoff. They hold on a *general class* of economies defined by (F1)–(F5) and by bounds on the slopes and curvature of the transition and on the rates of the payoffs. No derivative is assumed: each slope bound applies to every chord. The next state rises with the current state and with the sanction, the latter at a rate bounded away from zero. Both payoffs rise with the state at rates bounded away from zero, the target's the more so the heavier the sanction, because entrenchment softens a sanction's pain. A fixed version of this property is the parameter by which rent-seeking helps a target government decrease the costs of sanctions in Dorussen and Mo (2001, p. 407). The curvature of the sender's cost of sanctioning and the pain of a sanction to the target are bounded. The main text states these bounds in words where a result uses them; their symbols and precise forms are in Appendix B.1 of the Online Appendix. The paper's parametric examples, a quadratic–linear family of payoffs and a partial-adjustment law of motion, are in Appendix J.1 of the Online Appendix.

The positive lower bounds on the two payoffs' state rates give both incumbents a direct stake in the state. The sender's stake is the substantive assumption, and a reader may doubt that a sender gains from its adversary's entrenchment. The sender's own coalition is modeled as sustained by a visibly entrenched adversary, so the constituency the sanction strengthens abroad keeps the sanction popular at home. The assumption restates the sender-side interest-group logic of Kaempfer and Lowenberg (1988) as a continuation value. Sender-side politics of this kind are documented: in the case of Iran, lifting sanctions can carry a domestic political cost in the sanctioning country

[4]The Online Appendix lists every bound of the class in the form the proofs use (panel A0 of Appendix C.4) and gives the closed forms of the derived constants (Appendix N).

that imposing them does not (Bajoghli et al., 2024, p. 118). In a formal model of ending sanctions, audience costs lead the sender to prefer to remain with sanctions for domestic reasons as long as the target refuses to yield (Dorussen and Mo, 2001, p. 403). Three sender-side features are distinct. The bound on its state rate is a stake in the state itself. A state–sanction complementarity, where present, makes the return to sanctioning rise with the state, as a state–sanction cross term in the sender's payoff does; the audience-cost term of Dorussen and Mo (2001) is one, under which sanctions cost the sender less the more extreme the target's policy. And in the class the sender's one-period marginal return to a first unit of sanction against a non-compliant target is positive at every state, so its myopic best response sanctions a non-compliant target everywhere, and so does its forward-looking response whenever its continuation value does not fall with the state. Likewise the bound on the target's state rate describes the target incumbent's valuation of the organized influence the state measures, as the elite of Acemoglu and Robinson (2008, p. 268) invest in de facto power for the returns from controlling politics. An explicit interest-group polity, with two groups in each country, derives both stakes from a within-period political contest (Section 4).

Unlike the state, compliance pays the target only through the sender's response. It is costly at a fixed sanction, and it lowers the sender's marginal return to sanctioning: compliance and the sanction are substitutes in the sender's payoff, which has decreasing differences in them, at a rate we call the *substitution rate*. That concessions are costly to the target is 'a central notion in the literature on economic pressure' (Marinov, 2005, p. 567; see also Eaton and Engers, 1992, p. 903), and the sender of Tsebelis (1990, Assumptions 2 and 4) prefers to sanction a violation and not to sanction compliance, the binary form of a positive substitution rate. *Compliance neutrality*,

$$U_S(x, a, 0) = U_S(x, 0, 0), \tag{2}$$

says that compliance is payoff-irrelevant to the sender at a zero sanction: compliance matters to the sender only through its decision to sanction. It is not one of the paper's assumptions; the three results that construct relief state it (the relief side of Section 3.4, Proposition 11 and the sender-only case of Section 5.1). Along relief the sanction is zero, so the sender's value can be computed before the target's compliance is known; the positive results about non-compliance hold without it. Most coercion models let the sender value compliance directly (Eaton and Engers, 1992, p. 903; Marinov, 2005, p. 567); compliance neutrality follows instead the public-choice view of Kaempfer and Lowenberg (1988, eqs. (1) and (1′)), in which the payoffs in the sanctioning country are defined over the sanction and its domestic incidence rather than over the target's policy. For future reference, write $\underline{s}$ for the sanction that maximizes the sender's per-period payoff against full non-compliance at the state $x = 0$ (unique on the class, Section 2.3). A more entrenched adversary does not lower that sanction, so $\underline{s}$ is the least sanction a myopic sender imposes on a non-compliant target.

2.3 The myopic benchmark

We first analyze a benchmark in which neither government looks ahead; in this benchmark the target accommodates and sanctions end.[5] A myopic government gives no weight to the future political state. The sender's *myopic best response* is

$$\sigma^{\mathrm{M}}(x, a) \in \arg\max_{s \in [0,1]} U_S(x, a, s), \tag{3}$$

[5]The precise form of each assumption is in panel A0 of Appendix C.4 and in Appendix B of the Online Appendix, and Proposition 2 is proved in panel F of Appendix C.4.

that is, the sanction that gives the sender the highest payoff in the current period, at state x and compliance a. It exists by continuity on the compact interval and is unique on the class, so $\underline{s} = \sigma^{\mathrm{M}}(0,0)$: the myopic sanction against full non-compliance when entrenchment is zero. The class also requires it to be continuous in the state and compliance; only the results built on the myopic sender use this.

Definition 1 (Myopic equilibrium)**.** In the *myopic equilibrium* both discount factors are zero. The sender plays its myopic best response σ^{M} after every compliance level, and at every state the target chooses the compliance $\alpha(x)$ that maximizes its per-period payoff against that response, $U_T(x, a, \sigma^{\mathrm{M}}(x,a))$. Its sanction rule is $x \mapsto \sigma^{\mathrm{M}}(x, \alpha(x))$, and from an initial state x_0 its realized path is $s_t = \sigma^{\mathrm{M}}(x_t, \alpha(x_t))$, $x_{t+1} = F(x_t, s_t)$.

The benchmark uses three assumptions on the one-period problem. Each states one economic restriction and is marked with the results that use it; their dynamic counterparts, which count the future value of the state, come with the forward-looking governments in Section 2.4.

Assumption 1 (Myopic interiority)**.** In the one-period problem, the sender's marginal gain from sanctioning a non-compliant target falls short of the marginal cost of the maximal sanction, so a myopic sender never sanctions fully.

Used in: Proposition 2; Theorem 3 and Proposition 9, which compare with the benchmark; and the cases with a myopic sender in Section 5.1.
Interpretation: It keeps the myopic sender's sanction against a non-compliant target below the maximum. Interiority conditions of this kind are standard: Acemoglu and Robinson (2008, Assumption 3) impose a condition on primitives that makes the elite's investment in de facto power interior, and Bressan and Jiang (2020, Lemma 2.3) keep the follower's control in the interior of $[0,1]$ in a feedback Stackelberg game.

Assumption 2 (Compliance ends the myopic sanction)**.** In the one-period problem, full compliance removes the sender's marginal gain from sanctioning, so a myopic sender does not sanction a fully compliant target.

Also termed the *myopic lifting condition.*
Used in: Proposition 2, and through it in Theorem 3 and Proposition 9.
Interpretation: It makes relief available in exchange for compliance. It is the continuous-action form of the standard assumption that a sender prefers not to sanction a target that complies (Tsebelis, 1990, Assumption 2; Lacy and Niou, 2004, p. 31).

Assumption 3 (Myopic relief)**.** In the one-period problem, a unit of compliance costs the target less than the sanction pain it removes.

Used in: Proposition 2, and through it in Theorem 3 and Proposition 9; and in Proposition 6. In Theorem 12 it follows from the forward-looking relief condition (RL) of Section 5.1.
Interpretation: It makes relief worth its price. It is the marginal form of the familiar condition that sanctions succeed when the target's costs of non-compliance exceed its costs of compliance (Hufbauer et al., 2007, p. 50); in Marinov (2005, p. 566) the target leadership concedes only if the cost of noncompliance exceeds the cost of granting the sender's demands.

Proposition 2 (The myopic benchmark)**.** *For every economy in the class satisfying Assumptions 1–3, the myopic target's unique optimum at every state is partial compliance $a^{00}(x) \in (0,1)$—the minimum compliance level at which the myopic sanction falls to zero. Under this policy, the myopic equilibrium exhibits no sanctions at any state, and from any initial state the political state falls at least geometrically to zero.*

Without foresight, the target trades compliance for relief. Myopic interiority keeps the myopic sender's sanction against a non-compliant target below the maximum, and the myopic lifting condition causes it to vanish before compliance is complete. Thus a minimum compliance level $a^{00}(x)$ suffices to end the sanction. Myopic relief makes this relief worth the cost: in the one-period problem, a unit of compliance costs the target less than the pain it eliminates. The target complies exactly to this threshold, the sender imposes no sanctions, and in the absence of sanctions the state erodes at least geometrically—since the next-period state responds to the sanction at a rate bounded away from zero. Persistent sanctions therefore require a mechanism absent from the one-period framework: the value each government attaches to the political state its actions leave behind.

2.4 Forward-looking governments

We now allow both governments to look ahead: a forward-looking sender responds to observed compliance through its dynamic best response, and a forward-looking target optimizes given this response.[6] Given a sender continuation value V_S, the sender's dynamic objective at state x, with compliance a and sanction s, is

$$U_S(x, a, s) + \beta_S V_S(F(x, s)). \tag{4}$$

The first term is the sender's payoff $U_S(x, a, s)$ in the current period. The second term is the discounted value $\beta_S V_S(F(x, s))$ of the state that the sanction leaves for the next period, so the sender trades this period's payoff against the value of the state it leaves. The sender's *dynamic best response* $\sigma^{\text{dyn}}_{V_S}(x, a)$ is the largest sanction that maximizes this objective (the greatest-selection tie-breaking rule). This response exists because the argmax set is compact, and it is monotonic under the interval-dominance conditions applied below (Quah and Strulovici, 2009, Thm. 1). Strict concavity of the dynamic objective in the sanction level—including the continuation term—implies the argmax is a singleton. Given this response, the target's Bellman equation is

$$V_T(x) = \max_{a \in [0,1]} \left\{ U_T(x, a, \sigma^{\text{dyn}}_{V_S}(x, a)) + \beta_T V_T(F(x, \sigma^{\text{dyn}}_{V_S}(x, a))) \right\}. \tag{5}$$

Intuitively, the target weighs what compliance costs it today against two effects: compliance changes the sender's response $\sigma^{\text{dyn}}_{V_S}(x, a)$, and through that response it changes the state the target inherits next period. A target policy α is Bellman-optimal for V_T if $\alpha(x)$ attains the maximum on the right-hand side at every $x \in [0, 1]$.

The equilibrium concept builds on the dynamic best-response structure: the sender's response is optimal following every feasible compliance choice, both on and off the equilibrium path, and the target correctly anticipates this response. Strategies are *feedback* (or Markov) strategies: the target's compliance depends only on the current state, and the sender's sanction depends on the current state and the observed compliance level. These are Markov strategies in the sense of Maskin and Tirole (2001). The equilibrium defined below is the stationary infinite-horizon version of the feedback Stackelberg solution, which exhibits the Stackelberg property at every stage (Başar and Olsder, 1999, Def. 3.29 and Sec. 7.3). Breton et al. (1988) define analogous stationary strategies for discounted games, where the leader's strategy depends on the state and the follower's strategy depends on both the state and the leader's observed action. Formally, a *stationary feedback Stackelberg equilibrium* is a tuple $E = (V_T, V_S, \alpha)$ satisfying:

[6] The existence routes, the comparison with history-dependent strategies and the role of the myopic best response's continuity are in Appendix F and Appendix H.7 of the Online Appendix.

(E0) The target feedback policy $\alpha\colon [0,1] \to [0,1]$ is continuous.

(E1) V_T satisfies the target Bellman equation (5).

(E2) V_S satisfies the sender Bellman equation

$$V_S(x) = U_S\big(x, \alpha(x), \sigma_{V_S}^{\mathrm{dyn}}(x, \alpha(x))\big) + \beta_S V_S\Big(F\big(x, \sigma_{V_S}^{\mathrm{dyn}}(x, \alpha(x))\big)\Big).$$

That is, $V_S(x)$ is the sender's payoff now plus its discounted value from the next state, when the target follows α and the sender responds optimally.

(E3) α is Bellman-optimal for V_T.

(E4) The sender uses greatest-selection tie-breaking: $\sigma_{V_S}^{\mathrm{dyn}}$ is the largest maximizer of (4) over $s \in [0,1]$, pointwise in (x, a).

In models with complementarities, greatest-selection tie-breaking is the standard rule (Topkis, 1998, Thm. 2.8.3). Within the class of economies studied, the maximizer is unique, so this tie-breaking rule never binds. Let $\mathcal{E}$ denote the set of such equilibria. Because both value functions must solve their respective fixed-point equations, a sender solution given a fixed target policy is an intermediate construction step, not an equilibrium itself.

Each equilibrium generates a sanction rule and, from each initial state, a realized path of states and sanctions. Given $E \in \mathcal{E}$, the *sanction rule of* E is the map $x \mapsto \sigma_{V_S}^{\mathrm{dyn}}(x, \alpha(x))$. Given also an initial state $x_0 \in [0,1]$, the realized path of states and sanctions (x_t, s_t) is defined recursively by

$$s_t = \sigma_{V_S}^{\mathrm{dyn}}(x_t, \alpha(x_t)), \qquad x_{t+1} = F(x_t, s_t), \qquad x_0 \text{ given.} \tag{6}$$

In words, the rule sets the sanction from the current state, and the law of motion then moves the state. The map $x \mapsto F\big(x, \sigma_{V_S}^{\mathrm{dyn}}(x, \alpha(x))\big)$ is the *equilibrium law of motion*, and a *steady state* is a state that it leaves unchanged. The myopic equilibrium of Definition 1 is an equilibrium in $\mathcal{E}$ of the same economy with both discount factors set to zero: at a zero sender discount factor the dynamic best response is the myopic best response.

Assumptions for forward-looking results. Three assumptions underpin the benchmark comparison; two additional assumptions are used for specific results.[7] Assumption 4 is maintained throughout the class of economies; the rest are hypotheses of the results that invoke them. Conditions used only in a minority of results are stated alongside those results: compliance neutrality (Section 2.2), the strong entrenchment condition (Section 3.4), and the two conditions for the sender-only case (Section 5.1).

Assumption 4 (Concavity)**.** The curvature of the sender's sanctioning cost exceeds the curvature contributed by its continuation value: the marginal cost of sanctioning rises faster than the future gain from a more entrenched adversary can offset.

Used in: all results on the class, as a maintained class condition.
Interpretation: This condition guarantees the forward-looking sender has exactly one optimal sanction level at each state and compliance level. Strict concavity of the follower's stage-plus-continuation objective in its own action is the standard source of a unique follower response in

[7] The examples on either side of the entrenchment condition are in Appendix J.3 and Appendix B of the Online Appendix.

feedback Stackelberg games (Başar and Olsder, 1999, Remark 7.5). When the payoff is concave and the set of feasible choices is convex, standard recursive arguments imply concave continuation values (Stokey et al., 1989, Thm. 4.8), and stochastic games with complementarities restrict the transition so that the continuation term preserves the strict concavity of stage payoffs (Curtat, 1996, Thm. 4.1). The sender's value need not be globally concave, so Assumption 4 bounds the curvature it can add.

Assumption 5 (Entrenchment condition)**.** The target incumbent values the entrenchment generated by one unit of sanctions, in discounted terms, at least as much as the immediate pain of the sanction.

Formally: the minimum increase in next-period state per unit of sanction, multiplied by the minimum discounted value of a unit of state to the target under sanctions, weakly exceeds the maximum pain a unit of sanction can impose (inequality (B.2) in the Online Appendix).
Used in: Theorem 3 (to support non-compliance as the target's best reply); Proposition 9; persistence without hysteresis (Section 3.3); the participation result of Section 5.3; and the forward-looking target cases in Section 5.1 (including the long-run gap).

The entrenchment condition is a demanding statement about a government that would rather be sanctioned than weaken its base. It is consistent with the case-study descriptions of Section 5.3: in the model's terms, a leadership that can pass sanction costs to its citizens bears little pain from a sanction, and one reluctant to weaken its control values the power the state measures. The trade-off has precedents in political economy. In Acemoglu et al. (2012, Example 1) a sufficiently forward-looking elite blocks a reform that would raise its payoff, which a myopic elite adopts, because of the further changes the reform would engender; in Powell (2006, pp. 189–190) a faction in power prefers costly fighting to a settlement that lowers its chance of keeping power, the more so the lower the cost of fighting and the more valuable control of the state. For sanctions, non-democratic regimes can 'simply pass on the costs of the sanctions to the governed' (Nossal, quoted by Halcoussis et al., 2021, p. 156), and political elites 'mostly remain insulated' from the hardship (Peksen, 2021, p. 187). The condition describes a class of targets: sanctions raise the chance that a leader loses office on average (Marinov, 2005, p. 564), and the capacity to shift their costs is greater in single-party and military regimes than in personalist ones (Escribà-Folch and Wright, 2010, p. 341). It makes the forward-looking target maintain non-compliance, and it is sufficient, not necessary: an economy with quadratic–linear payoffs fails it and still has non-compliance as the target's unique optimum. Economies that meet the conditions exist inside the quadratic–linear family and outside it, including one whose transition is affine in neither coordinate.

Assumption 6 (Increasing differences)**.** The sender's dynamic objective exhibits increasing differences in the sanction and the state: a more entrenched adversary raises the marginal attractiveness of sanctioning for the sender by enough to offset two opposing forward-looking forces: the curvature of the sender's continuation value, and the state-dependent effect of sanctions on next-period state dynamics.

Used in: Theorem 3; persistence without hysteresis (Section 3.3). It ensures the sender's response to non-compliance is non-decreasing in the state; Proposition 9, Proposition 7, the long-run gap of Section 5.1, and the participation result of Section 5.3 rely only on this monotonicity property.
Interpretation: The condition holds when the transition follows a partial-adjustment law—shifting the state a fixed fraction of the way toward the current sanction—and the sender's payoff is convex in the state. Increasing differences in a player's own action and the state is the standard sufficient condition for monotone state-contingent policies (Topkis, 1998, Thm. 3.9.2; Curtat, 1996; Balbus

et al., 2014, p. 826). Standard regularity conditions on transition rules exclude deterministic laws of motion (Amir, 2002, p. 44), so the assumption is imposed on the sender's dynamic objective. As in Hopenhayn and Prescott (1992, p. 1403), the complementarity must be large relative to competing curvature effects; under the partial-adjustment law, the continuation term exhibits increasing differences if and only if the continuation value is convex (Topkis, 1998, Lemma 2.6.2).

Assumption 7 (Interiority)**.** Even accounting for the future value of a higher state, the sender's marginal gain from sanctioning a non-compliant target falls short of the marginal cost of the maximum sanction. Thus a forward-looking sender never imposes full sanctions.

Used in: the premium bound of Section 3.2; the result in Theorem 3 that forward-looking sanctions are never at full intensity; Proposition 6; the results of Section 3.3 on full-non-compliance equilibria and on persistence without hysteresis; the relief-side analysis of Section 3.4; and Theorem 12.
Interpretation: This condition implies Assumption 1, and it allows a forward-looking sender to sanction strictly more heavily than a myopic one without reaching the upper bound.

Assumption 8 (Compliance ends the sanction)**.** Full compliance eliminates the sender's marginal gain from sanctioning, including future gains, so a forward-looking sender also does not sanction a fully compliant target.

Also termed the *lifting condition*.
Used in: Proposition 6; the results of Section 3.3 on full-non-compliance equilibria and on persistence without hysteresis; the relief-side analysis of Section 3.4; and the myopic-target cases in Section 5.1.
Interpretation: This condition implies Assumption 2, and it preserves the availability of relief in exchange for compliance under forward-looking senders. In dynamic sanction games, compliance ends sanctions by the rules of the game (Eaton and Engers, 1992, p. 914); here it arises endogenously from the forward-looking sender's incentives.

3 Equilibrium analysis: foresight and sanction persistence

This section compares forward-looking governments with the myopic benchmark. Section 3.1 shows that foresight turns relief into non-compliance and keeps sanctions and the political state positive. Section 3.2 bounds the sender's part of that difference. Section 3.3 gives conditions under which a high political state stays high, Section 3.4 asks how patient the target must be and collects two further consequences of non-compliance. Each subsection opens with a footnote that says where its proofs are. Existence, uniqueness within a stated set and value bounds are kept separate, each under the hypotheses of its own result.

3.1 Comparison with the myopic benchmark

We now turn to the main result: when both governments are forward-looking, the same economy supports an equilibrium with persistent non-compliance and indefinite sanctions.[8] Stage payoffs, feasible sets, and the transition law are identical to the benchmark; only the discount factors are positive. Superscripts index the case: 00 for the myopic benchmark, 11 for the case with two forward-looking governments. The non-compliance equilibrium is constructed in two steps. First, the sender's value under full non-compliance is derived as the fixed point of its Bellman operator; at every state, the forward-looking sanction is then strictly higher than the corresponding myopic

[8] Theorem 3 is proved in panel F of Appendix C.4 of the Online Appendix, and the full-non-compliance equilibrium it uses (Proposition B.1) is constructed in the same appendix; the component statements are in Appendix B.

Table 1: Notation, by family in the order the text introduces them. Symbols that appear in a single result are defined there.

Symbol	Meaning
The game (Section 2.1)	
x_t, a_t, s_t	political state, target compliance, sanction at date t, each in $[0,1]$
β_T, β_S	discount factors of the target and the sender
F	law of motion of the state, $x_{t+1} = F(x_t, s_t)$
Payoffs, values and policies (Section 2.2)	
U_T, U_S	per-period payoffs of the target and the sender governments
V_T, V_S	their continuation values
α	target policy (compliance as a function of the state)
$\underline{s}$	least sanction a myopic sender imposes on a non-compliant target
The myopic benchmark (Section 2.3)	
σ^{M}	sender's myopic best response
a^{00}	least compliance that ends the myopic sanction (Proposition 2)
Forward-looking governments (Section 2.4)	
$\sigma_{V_S}^{\mathrm{dyn}}$	sender's dynamic best response
$\mathcal{E}$	set of stationary feedback Stackelberg equilibria
Equilibrium outcomes (Section 3)	
x_∞^{11}	steady state of the forward-looking equilibrium (Theorem 3); superscripts name the case
$\bar{x}$	threshold of a high basin $[\bar{x}, 1]$ (Proposition 5)
β_T^*	threshold discount factor for full non-compliance (Proposition 6)
Welfare (Section 4)	
W_g, θ_g	welfare of interest group g and its political weight (Definition 8)
W	aggregate welfare, the sum of the W_g
One government forward-looking (Section 5.1)	
a^{01}	least compliance that ends a forward-looking sender's sanction (Table 3)
x_∞^{10}	long-run state when a forward-looking target faces a myopic sender (Theorem 12)

sanction $\sigma^{\mathrm{M}}(x, 0)$, the sanction a myopic sender sets at state x against full non-compliance. Second, under the entrenchment and increasing-differences conditions, full non-compliance is the target's unique best reply to that response, and the two value functions paired with the full-non-compliance policy constitute an equilibrium in $\mathcal{E}$. Recall that $\mathcal{E}$ is the set of stationary feedback Stackelberg equilibria (Section 2.4) and that $\underline{s}$ is the least sanction a myopic sender imposes on a non-compliant target (Section 2.2).

Theorem 3 (Foresight vs. the myopic benchmark)**.** *For every economy in the class satisfying Assumptions 1–6, there exists an equilibrium in $\mathcal{E}$ such that:*

(i) the target maintains full non-compliance at every state, and the sender's sanction level lies strictly above the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$;

(ii) from every initial state the amount by which the political state falls short of $\underline{s}$ shrinks at least geometrically; starting from a common initial state, the state lies strictly above the path of the myopic benchmark (Proposition 2) at every positive date;

(iii) the state converges to a steady state $x^{11}_{\infty} > \underline{s}$, at which the sanction level equals the state level.

Table 2: The myopic benchmark and the forward-looking equilibrium of one economy: compliance, sanction and the limit of the political state. Stage payoffs, feasible actions and the transition are the same in both columns; the conditions of Theorem 3 are maintained.

	myopic governments (Proposition 2)	forward-looking governments (Theorem 3)
compliance	$a^{00}(x)$, the least that ends the myopic sanction	full non-compliance (compliance 0)
sanction	zero at every state	strictly above the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$, and below 1 under interiority
political state	falls to 0, at least geometrically	above the benchmark's path at every positive date; converges to a steady state above $\underline{s}$, which may depend on the start

Foresight shifts outcomes from relief to non-compliance, and under non-compliance both sanctions and the political state remain positive. Conclusion (ii) implies that a state that starts below $\underline{s}$ rises toward $\underline{s}$ or above it, because its shortfall from $\underline{s}$ shrinks at least geometrically. Conclusion (iii) says that x^{11}_{∞}, the long-run political state when both governments look ahead, lies above $\underline{s}$; at this state the sanction equals the state, so by (F4) the state stays there. In the myopic benchmark of Proposition 2, the target accommodates and the state erodes to zero (Table 2). The forward-looking target instead chooses non-compliance because, under the entrenchment condition, the entrenchment generated by sanctions is worth at least as much as the immediate pain they impose, making it unprofitable to trade compliance for relief. For its part, the forward-looking sender sanctions a non-compliant target more heavily than a myopic sender would. The state then converges to a positive level from any initial condition, declining toward it from sufficiently high starting points. Under interiority (Assumption 7), the sender never imposes full sanctions. Conclusions (i)–(iii) hold at every equilibrium in $\mathcal{E}$ with full target non-compliance—not only at the constructed equilibrium—because all such equilibria share the same constructed values and sanction rule. The steady-state limit may depend on initial conditions unless the equilibrium law of motion is contractive in state distances (Section 3.3).

The theorem changes both discount factors at once, holding every other primitive fixed. It shows that foresight makes sanctions persist; it does not by itself say which government's foresight

does what. Section 3.2 bounds the sender's part of the difference, and Section 5.1 solves the cases in which only one government looks forward.

3.2 The sender's foresight premium

The comparison so far concerns the equilibrium as a whole; we next isolate the sender's contribution, the foresight premium: the amount by which the sender's forward-looking sanction exceeds its myopic sanction at a given compliance level. This premium has a closed-form lower bound at full non-compliance that increases with the sender's stake in the political state.[9] Under interiority, at every state the forward-looking sender's sanction under non-compliance exceeds the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$ (the myopic sender's sanction against full non-compliance) by at least a strictly positive closed-form bound. The numerator captures the minimum increase in the sender's discounted value from a unit of sanctions via the next-period state: the sender's per-period stake in the state, discounted over the state's persistence rate, multiplied by the minimum response of the next-period state to sanctions. The denominator—the maximum curvature of the sender's sanctioning cost function—converts this marginal gain into a sanction increase. The bound rises with the sender's discount factor, its stake in the state, and the responsiveness of the transition to sanctions; its general form is given by (B.3) in the Online Appendix. These lower bounds hold for fixed primitives. The bound is attained at one example economy, so its coefficient cannot be uniformly scaled up across the class. The foresight premium rests on a comparison that holds broadly outside the class. Recall that σ^{M} is the myopic best response and $\sigma^{\mathrm{dyn}}_{V_S}$ the dynamic best response, which also counts the value V_S of the next state; α is a target policy.

Theorem 4 (Dynamic vs. myopic sanctions at a fixed target policy)**.** *Let V_S be a bounded continuous sender continuation value that is non-decreasing in the political state. For any target policy $\alpha\colon [0,1] \to [0,1]$ and every state $x \in [0,1]$,*

$$\sigma^{\mathrm{M}}(x, \alpha(x)) \leq \sigma^{\mathrm{dyn}}_{V_S}(x, \alpha(x)). \tag{7}$$

In particular, inequality (7) *holds at every equilibrium $E = (V_T, V_S, \alpha) \in \mathcal{E}$ whose sender value is non-decreasing in the state.*

The intuition relies on monotonicity: at a fixed target policy, a higher sanction leaves a weakly higher successor state, so a non-decreasing continuation value can only increase the largest of the optimal sanctions. Inequality (7) thus implies that a sender that values a more entrenched adversary never sanctions less than a myopic sender facing the same compliance. The comparison holds under general transition rules and requires neither an equilibrium nor restriction to high states. Every stage maximizer, not only the selected one, lies weakly below the forward-looking sanction. The inequality becomes strict when no sanction below the greatest stage maximizer yields a higher continuation value (automatic when the continuation value is non-decreasing) and, for some sanction level, the discounted continuation gain from moving to that level outweighs the stage loss relative to the greatest stage maximizer. This version requires no concavity, differentiability, or positive discount.

Relative to the myopic equilibrium at its own target policy, the forward-looking equilibrium sanctions at least as heavily at every state when its sender value is non-decreasing, its target complies no more, and the sender's myopic best response does not increase with compliance. The difference operates through two separate channels at each state where the forward-looking target

[9] The bound and the attainment example are proved in Appendix C.5 of the Online Appendix; the comparison at a fixed target policy and its strict form in Appendix C.2.

complies no more than the myopic one: lower target compliance can only raise the myopic best response (the indirect effect), and sender foresight can only raise sanctions at that compliance level (the direct effect, Theorem 4). At the level of the sanction rule, these two effects—the target's part and the sender's part—sum to the total difference between Proposition 2 and Theorem 3.

3.3 Sanction lock-in and history dependence

The results so far describe constructed equilibria. We now show that, under the conditions stated below, persistence sustains itself once it starts, a property we call lock-in: once the political state reaches a high level, equilibrium play keeps it there, the state never exits a high basin, and sanctions never decline.[10] Whereas the non-compliance result focuses on the target side, lock-in incorporates the sender's incentives. Under the conditions below, raising sanctions to match the state is profitable for the sender across the entire high basin—from a threshold $\bar{x}$ up to the upper bound—and a sanction at or above the current state prevents the state from falling. The result requires none of the class-wide bounds: it holds at every equilibrium of any economy satisfying its conditions. A temporary shock—a finite period of exogenously set sanctions—can move the state into the high basin. In the proposition, $E = (V_T, V_S, \alpha)$ is an equilibrium: the two governments' value functions and the target's policy.

Proposition 5 (Lock-in)**.** *Let $E = (V_T, V_S, \alpha) \in \mathcal{E}$ and let $[\bar{x}, 1]$ be a high basin of states. Suppose that the equilibrium value functions lie within closed sets of bounded continuous functions, with sender values Lipschitz-continuous in the state, and that the target policy is the one induced by these value functions. Suppose further that, at every pair of values in those sets (with the induced target policy substituted):*

(a) monotonicity*: on the high basin, the sender's objective satisfies the interval dominance order in the sanction and the state (Quah and Strulovici, 2009), so the largest of its optimal sanctions rises with the state; and*

(b) level condition*: at every state in the basin, raising sanctions from any lower level up to the current state level increases the sender's current payoff by more than the maximum possible change in its discounted continuation value.*

Then from every initial state $x_0 \in [\bar{x}, 1]$, the equilibrium path $s_t = \sigma_{V_S}^{\mathrm{dyn}}(x_t, \alpha(x_t))$ and $x_{t+1} = F(x_t, s_t)$ satisfies for all $t \geq 0$:

(i) $x_t \in [\bar{x}, 1]$*;*

(ii) $s_t \geq \bar{x}$*;*

(iii) $s_t \leq s_{t+1}$*.*

If in addition the transition satisfies (F1)–(F4) *and full sanctions raise every state below the basin, then from any initial state a temporary shock holding sanctions at the maximum level for a sufficient duration moves the state into $[\bar{x}, 1]$ once the shock ends, and properties* (i)–(iii) *hold along the equilibrium path of E restarted from that state.*

[10] Proposition 5 restates the persistence and temporary-shock theorems of Appendix D in the Online Appendix, which also gives their value sets, the constants of the level condition and the shock length; the low-basin counterpart is Theorem D.5, and an equilibrium with both basins in Appendix K.2.

Property (i) makes the high basin an absorbing region of the equilibrium path. In words, (i) the state never falls below $\bar{x}$; (ii) the sanction never falls below $\bar{x}$; (iii) the sanction never falls from one date to the next. Democracy can become an absorbing state in the same sense in Acemoglu and Robinson (2008, Corollary 2, p. 279), when investment in de facto power is no longer always profitable for the elite. The entry result gives a temporary event a lasting effect, as in Acemoglu and Robinson (2001, p. 939), where a transitory threat of revolution forces the elite to extend the franchise; there the lasting change runs through the credibility of concessions, here through the political state that the shock moves.

The conclusions of Proposition 5 rely on different subsets of its conditions. For economies in the class satisfying interiority and the lifting condition, a weaker form of lock-in holds at every full-non-compliance equilibrium. The set-based conditions restrict equilibrium value functions to sets of bounded functions with Lipschitz sender values, and require the target policy to be the one induced by the value pair. The monotonicity condition ensures the sender's response (with the target policy substituted) is non-decreasing in the state across the basin. The level condition requires that, at each basin state, raising sanctions from any lower level up to the state increases the sender's current payoff by more than any associated change in discounted continuation value. Both conditions are imposed pointwise across all value pairs in the sets. The no-exit properties (i) and (ii) rely only on the set conditions and the level condition; the non-decreasing sanction property (iii) additionally requires the monotonicity condition. No assumption on how the target ranks sanction levels is needed. The entry result additionally requires conditions (F1)–(F4) and that full sanctions raise every state below $\bar{x}$; the Online Appendix provides an explicit sufficient shock duration.

For the class of economies satisfying interiority and the lifting condition, at every equilibrium with full target non-compliance, once a temporary shock moves the state to or above $\underline{s}$ (the least sanction a myopic sender imposes on a non-compliant target), the state remains there and sanctions never fall below $\underline{s}$.

The argument also runs in reverse: the political state can entrench low sanctions as well as high ones. Under low-basin versions of the conditions, the state remains in the basin and realized sanctions never rise for any starting point within a low basin. When a single equilibrium satisfies the sender's conditions on both a high basin and a disjoint low basin below it, the initial state selects between the two basins within that equilibrium. Persistence can also hold independently of initial conditions: for economies in the class satisfying interiority, the lifting condition, and the entrenchment and increasing-differences conditions, and whose equilibrium law of motion has an explicit Lipschitz constant strictly below one, the constructed equilibrium has exactly one steady state. Every initial state converges to this state, and temporary sanction shocks have no permanent effect on the long-run limit.

The reverse experiment—a temporary suspension of sanctions—can reduce the state below a given level within a computable time horizon.[11] After entry into a high basin, equilibrium play generates no endogenous exit, but relief remains mechanically feasible: the transition law still erodes the state under low sanctions. Whether de-escalation persists then depends on the resumed equilibrium: it persists if the equilibrium's sanction rule remains at or below the state level for states below that threshold. Entry and exit durations derive from different segments of the same transition function, so they need not be equal. For any transition symmetric under reflection of the unit square, $F(1-x, 1-s) = 1-F(x, s)$ (the state moves up toward a sanction above it in the same way as it moves down toward one below it), such as the partial-adjustment law, the exit-duration

[11] Entry and exit durations, persistence after relief, the reading of the lock-in conditions and the example economies that meet them are in Appendices D and J of the Online Appendix.

bound (before rounding to integer periods) exceeds the entry-duration bound exactly when that level lies below one half.

3.4 Target patience and non-compliance equilibria

The analysis so far compares a myopic with a forward-looking target. A natural question is how patient the target must be for non-compliance to prevail; the comparison extends across discount factors and collapses to a single threshold in the linear–quadratic special case.[12] In this special case: the transition responds to sanctions at a constant rate, which combined with $F(x, x) = x$ makes it affine; the curvature of the sender's sanctioning cost is constant; the substitution rate, the target's compliance cost, its sanction pain, and its valuation of the state are all constant; the target's payoff has no state–sanction cross term; and the curvature bounds on the sender's value and its state-dependent payoff are zero. The target's *net cost* of sanctions equals their pain minus the compliance cost of ending them.

Proposition 6 (Threshold discount factor for full non-compliance)**.** *In the linear–quadratic special case, for an economy satisfying interiority, the myopic relief condition, and the lifting condition, there exists a threshold discount factor β_T^* (in closed form in the Online Appendix) such that an equilibrium in $\mathcal{E}$ with full non-compliance at every state exists if and only if $\beta_T \geq \beta_T^*$. When $\beta_T^* < 1$, these discount factors span the interval $[\beta_T^*, 1)$.*

The threshold balances the target's net cost of sanctions against the value of the entrenchment they build. In words, an equilibrium with full non-compliance exists exactly when the target's discount factor β_T is at least β_T^*. It rises with this net cost and falls with the target's valuation of the political state. The non-existence result applies across all equilibria in $\mathcal{E}$: below β_T^*, no equilibrium exhibits full non-compliance. When the net cost equals or exceeds the target's valuation of the political state, $\beta_T^* \geq 1$ and no discount factor can support full non-compliance. The sender's patience changes the level of its response, and thus the compliance threshold for relief, but in the special case it does not shift the target's threshold, which depends on the target's costs, pain, state valuation, and discount factor. Acemoglu and Jensen (2015, Lemma 2, Proposition 1) derive robust discount-factor comparative statics for single-agent dynamic programs and for aggregates of price-taking agents; our result concerns one government's patience in a two-player strategic setting.

Outside the special case, the sufficient conditions for relief and for non-compliance need not coincide at a single discount factor, but a more patient target still preserves the full-non-compliance equilibrium. The target's *slope condition* at a non-negative slope is the condition under which its value under full non-compliance rises with the state at least at that slope. It requires that the target's per-period stake under sanctions cover the slope effect—net of state persistence—plus the portion of sanction pain left uncompensated by entrenchment at that slope, scaled by the responsiveness of the sender's reaction to changes in the state. Under the entrenchment condition, the slope condition holds at the slope that the entrenchment condition uses.

Proposition 7 (Greater target patience preserves the full-non-compliance equilibrium)**.** *Consider two economies in the class that differ only in the target's discount factor, with $\beta_T \leq \beta_T'$. Suppose the sender's response at full non-compliance at its constructed value is non-decreasing in the state, and the slope condition holds at some non-negative slope for the less patient economy. Then:*

[12] The threshold, the general sufficient conditions for relief and for non-compliance, the exact thresholds and the inheritance by a more patient target are proved in Appendices C.6 and C.7 of the Online Appendix, and branch invariance in Appendix C.5; the general conditions, the precise conditions of the special case and the closed form of the threshold are stated in Appendix B.5.

(i) the slope condition holds at the same slope for the more patient economy; and

(ii) if an equilibrium in $\mathcal{E}$ with full non-compliance at every state exists at β_T, one exists at β'_T.

Under these two conditions, if full non-compliance is an equilibrium at a given target discount factor, it remains an equilibrium at every higher discount factor. This result extends to discount factors for which neither sufficient condition holds. The sender's value function and best response at full non-compliance are identical across both economies. Greater target patience increases the value of the entrenchment that compliance would forgo, since compliance lowers the sender's response; the net gain from non-compliance is therefore preserved. The slope condition ensures that the target's value under full non-compliance is non-decreasing in the state, as required for the argument; assuming this monotonicity directly at either economy also suffices. The result is a comparative statics result for the viability of a single profile—full non-compliance at all states—as an equilibrium.

Across the full class of economies, sufficient conditions on the target's discount factor for relief and for non-compliance, their exact thresholds, and the result that patience—like every other target parameter—shapes whether a full-non-compliance equilibrium exists but not its equilibrium trajectory are all stated in Appendix B.5 of the Online Appendix.

Under the strong entrenchment condition, non-compliance also bounds how far the sender can lower the target's payoff and, across the class, pins down every equilibrium with a non-decreasing sanction rule.[13] The *strong entrenchment condition* strengthens Assumption 5 by removing the offset from entrenchment's softening of the pain of additional sanctions, making it a more demanding requirement; only the two results in this paragraph rely on it. Permanent relief—with the target maintaining non-compliance throughout—then constitutes the harshest treatment the sender can impose: the resulting payoff equals the target's minmax payoff in pure strategies. The target can guarantee itself this payoff by maintaining non-compliance in every period against any history-dependent sanction strategy, and every equilibrium in $\mathcal{E}$ yields the target at least this payoff. History-dependent strategies of this form, together with a random signal that all players observe, support the sequential equilibria of Phelan and Stacchetti (2001), and the minmax payoff is the standard object through which the conflict literature characterizes all-equilibrium results (Powell, 2004; Krainin and Wiseman, 2016). The intuition is that, under the strong form, the entrenchment built by sanctions is worth at least as much to the target as the associated pain. Under the same condition, across the class, every equilibrium with a sanction rule non-decreasing in the state exhibits full non-compliance, along with the constructed values and response; no equilibrium exhibits zero sanctions at every state. The next section turns from the governments' incentives to welfare.

4 Welfare analysis

4.1 Interest-group framework and aggregate welfare

We now turn to welfare. Because government payoffs weight the interest groups to which each government is accountable, the game alone does not pin down a welfare ranking.[14] Even insider economic interests can gain from relief. In Iran, listed firms affiliated with the conglomerates of the Revolutionary Guard and the Supreme Leader gained more than other listed firms following news of progress toward sanctions relief in 2012–2015 (Draca et al., 2023). This evidence concerns the economic interests of insiders, not the political state variable, which captures their political power. Welfare analysis therefore requires its own set of primitives.

[13] Both results are stated in Appendix B and proved in Appendix C.7 of the Online Appendix.

[14] The interest-group polity's algebra and its sign conditions are in Appendix L of the Online Appendix.

Definition 8 (Interest-group representation)**.** An *interest-group representation* of government payoffs consists of finite sets of sender groups and target groups, and, for each group g, a welfare function $W_g(x, a, s)$ and a political weight $\theta_g(x, a, s)$—positive at every feasible state and action—such that for every state and action

$$U_S = \sum_{g \in \text{Sender}} \theta_g W_g, \qquad U_T = \sum_{g \in \text{Target}} \theta_g W_g.$$

That is, each government's payoff is a weighted sum of the welfare of the groups it answers to, and θ_g is the weight of group g. *Aggregate welfare* is defined as $W = \sum_g W_g$, with every group weighted equally. A *planner* ranks paths of states and actions by discounted aggregate welfare, using a discount factor $\beta \in [0, 1)$ specific to the planner.

Group welfare functions are unrestricted; political weights follow the standard tradition of Grossman and Helpman (1994) and Becker (1983), deriving from each group's political pressure. At every state and action,

$$U_S + U_T = W + \sum_g (\theta_g - 1) W_g. \tag{8}$$

Thus the sum of the two governments' payoffs deviates from aggregate welfare by the sum over groups of each group's welfare multiplied by its political weight minus one. The interest-group polity of Kaempfer and Lowenberg (1988)—with a pro-sanction group and a commercial lobby in the sender country, and a regime-aligned group and an opposition in the target country—constitutes an economically structured representation: four sign conditions on its coefficients deliver the game's stage assumptions.

4.2 Indeterminacy of welfare rankings

The source of this indeterminacy is simple: a government's payoff pins down only the weighted sum of its groups' welfare, not the welfare of each individual group.[15] Multiplying one group's welfare by a positive factor while dividing its political weight by the same factor leaves government payoffs—and thus the game and its equilibria—unchanged, and changes aggregate welfare by exactly the change in that group's welfare (Proposition C.15 of the Online Appendix). If a group is strictly better off along one path than another by a uniform margin, and welfare is bounded along both paths, there exists some scaling factor that makes the planner rank the first path strictly above the second at every planner discount factor.

When applied to the myopic benchmark and the forward-looking equilibrium, this same degree of freedom reverses their welfare ranking.

Proposition 9 (Welfare non-identification on the class)**.** *Consider an economy in the class satisfying Assumptions 1–5, and for which the sender's response at full non-compliance at its constructed value is non-decreasing in the state, as under Assumption 6. There exist two interest-group representations of its government payoffs—each with two groups per country and positive constant weights—such that, at every initial state and every common planner discount factor $\beta \in [0, 1)$, discounted aggregate welfare ranks the forward-looking equilibrium of Theorem 3 strictly above the myopic benchmark of Proposition 2 under one representation, and strictly below it under the other.*

Both rankings are feasible because government payoffs impose no restriction on how valuable sanctions are to the underlying groups. The two representations assign to sender groups a net gain

[15] The proofs of Propositions 9 and C.15 are in Appendix C.9 of the Online Appendix.

per unit of sanctions of either sign, and make compliance a transfer between groups within the target country. The benchmark involves no sanctions, and the forward-looking equilibrium imposes sanctions of at least $\underline{s}$ (the least sanction a myopic sender imposes on a non-compliant target) in every period, so a sufficiently large net gain of either sign orders the two welfare flows at every date. Both representations are accounting constructs. Two economically grounded examples are provided in Appendix L of the Online Appendix: in the Kaempfer–Lowenberg polity, two adjacent polities with the same game structure reverse the ranking of zero sanctions versus full sanctions; within the quadratic–linear family, linear microfoundations reverse the ranking of the four cases (Proposition L.5). This ranking reversal extends to the four cases analyzed in Section 5.1.

4.3 Optimal policy and persistent sanctions

We next ask when a social planner would impose sanctions at all: whether zero sanctions are socially optimal depends on whether the first unit of sanctions yields positive net social returns.[16]

Proposition 10 (When zero sanctions are socially optimal)**.** *Let W be a per-period welfare function bounded over all feasible states and actions, and let $\beta \in [0,1)$ be the planner's discount factor. Suppose there exist a compliance rule $\hat{\alpha}$ and constants η, B, $\lambda \geq 0$ and L such that, for all feasible states $x \leq x'$, compliance levels a, and sanction levels s,*

(i) $W(x,a,s) \leq W(x,\hat{\alpha}(x),0) + \eta\, s$: at zero sanctions, the compliance rule $\hat{\alpha}(x)$ maximizes welfare, and sanctions add at most η per unit;

(ii) $W(x',\hat{\alpha}(x'),0) - W(x,\hat{\alpha}(x),0) \leq B\,(x'-x)$: at zero sanctions, welfare rises with the state at most at rate B;

(iii) $F(x',0) - F(x,0) \leq \lambda\,(x'-x)$ and $F(x,s) - F(x,0) \leq L\,s$: without sanctions the state persists at most at rate λ, and a unit of sanction raises the next state by at most L;

and that

$$\eta + \beta\,\frac{\max\{B,0\}}{1-\beta\lambda}\,L \leq 0, \qquad \beta\lambda < 1.$$

Then the path with zero sanctions and compliance rule $\hat{\alpha}$ achieves the planner's optimum against every time-dependent plan over both actions, from every feasible starting point.

The condition requires that the first unit of sanctions yields non-positive net social returns. The return has two parts. The direct part is η, the social return of a unit of sanctions within the period. The indirect part, $\beta \max\{B,0\}L/(1-\beta\lambda)$, is the most that the higher state left by sanctions can add: a unit of sanction raises the next state by at most L, the state persists at a rate of at most λ, each unit of state adds at most $\max\{B,0\}$ to welfare per period, and the planner discounts at β. Their sum must be non-positive. When welfare under zero sanctions does not increase with the state ($B \leq 0$), the condition reduces to $\eta \leq 0$. The monotonicity of the transition law, (F2)–(F3), is the only model property the proposition uses. In the Kaempfer–Lowenberg polity, η equals the sender groups' net gain from a unit of sanctions minus the target's direct cost, and B equals the difference between the two target groups' valuations of the state (Proposition L.3). For one example economy of that polity, equilibrium play supports full sanctions against full non-compliance from every starting point, while zero sanctions achieve the planner's optimum, and equilibrium welfare falls strictly short of the optimum at every starting point and every planner discount factor.

Persistence and efficiency are distinct questions. A welfare ranking of constructed equilibria compares equilibrium trajectories; efficiency compares equilibrium play to every feasible path.

[16]The proofs of Propositions 10 and 11 are in Appendix C.9 of the Online Appendix.

Proposition 11 (Persistent play can be socially optimal)**.** *Consider an economy in the class satisfying compliance neutrality* (2)*. There exists an interest-group representation—with two groups per country and positive constant weights—whose aggregate welfare coincides with the sender's payoff U_S, such that for every equilibrium in $\mathcal{E}$ with full target non-compliance, at the planner discount factor β_S, equilibrium play achieves the planner's optimum against every time-dependent plan over both actions, from every feasible starting point.*

Compliance reduces the sender's marginal return to sanctioning, and under compliance neutrality it has no value to the sender in the absence of sanctions, so compliance never increases the sender's payoff. A planner who evaluates welfare using the sender's payoff at the sender's discount factor therefore faces the same problem as the sender against full non-compliance, and equilibrium play solves it. Under the entrenchment condition and a sender response at full non-compliance that is non-decreasing in the state, the forward-looking equilibrium of Theorem 3 satisfies this property. For the quadratic–linear family, the corresponding result for specifications without state–sanction cross terms admits a witness with linear microfoundations (Corollary L.6). The discussion first solves the cases in which only one government looks forward, and then asks whether a persistent sanction is permanent.

5 Extensions and discussion

5.1 Single-sided foresight cases

We have so far compared an economy in which neither government looks ahead with one in which both do; solving the two cases in which only one government is forward-looking attributes the difference between them to each government separately.[17] The same economy can be analyzed with either discount factor set to zero. Stage payoffs, feasible sets, and the transition law remain identical, and both governments optimize fully in every case. The first digit of a case label indexes whether the target is forward-looking, the second whether the sender is: 00 is the myopic benchmark, 11 the forward-looking case of Theorem 3, 10 the target-only case, and 01 the sender-only case (Table 3). Neighboring cases differ in exactly one discount factor, so the gap between their constructed equilibria is attributable to that government's foresight. The sender-only case relies on two additional conditions that no other case uses, and which are therefore not among the paper's core assumptions; they are described next.

First, *semiconvexity in the sanction* (T6): the next state curves only boundedly as a function of sanction intensity. As a function of sanctions, the transition function lies above its chords by a uniformly bounded amount. This property caps the curvature of the forward-looking sender's objective, and with it the level of compliance needed to reduce the sender's sanction; the relief-side analysis of Section 3.4 also invokes this condition. Semiconvexity with a linear modulus is a standard concept from Cannarsa and Sinestrari (2004, Prop. 1.1.3 and Def. 2.1.1); bounds of this form on dynamics—which vanish for affine laws—are the standard regularity conditions under which curvature bounds extend through dynamic-programming operators (Bardi and Capuzzo-Dolcetta, 1997, Thm. III.3.26).

Second, the *forward-looking relief condition* (RL): against a forward-looking sender, a unit of compliance costs the target less than the sanction pain it eliminates. The sanction reduction is

[17]The component results (the sender's response under interiority and the lifting condition, and the target-only and sender-only cases) are stated in Appendix B of the Online Appendix and proved in Appendix C.4; the foresight premium along paths and the long-run gap in Appendix C.5; the welfare reversal across the four cases in Appendix C.9, and the partial identification on the family in Appendix L.

measured by the curvature of the sender's forward-looking objective. A forward-looking sender demands more compliance for relief, and (RL) implies the myopic relief condition (Assumption 3). Its counterpart in Eaton and Engers (1992, p. 915) is that sanctions be 'minimally effective', so that the target prefers complying to suffering them permanently. The precise forms of both conditions appear in Appendix C.4 of the Online Appendix.

Table 3: Constructed equilibria in the four cases of one economy (the target's discount factor by row, the sender's by column): compliance, sanction and the limit of the political state. Stage payoffs, feasible actions and the transition are identical across cells; the four-case conditions (Assumptions 5–8, compliance neutrality, and (T6) and (RL) above) are maintained. Each cell describes the constructed equilibrium of that case.

target \ sender	myopic sender	forward-looking sender
myopic target	compliance $a^{00}(x)$, the least that ends the myopic sanction, brings the sanction to zero; the state converges to 0.	compliance $a^{01}(x) > a^{00}(x)$, the least that brings the forward-looking sender's sanction to zero, which can be full compliance; the state converges to 0.
forward-looking target	full non-compliance (compliance 0); the sanction is the myopic sanction $\sigma^{\mathrm{M}}(x, 0)$; the state converges to a positive fixed point x^{10}_{∞} of the myopic sanction rule.	full non-compliance (compliance 0); the sanction lies strictly between the myopic sanction and 1; the state converges to a level strictly above x^{10}_{∞}, which may depend on the start.

Theorem 12 (Four cases)**.** *For every economy in the class satisfying the four-case conditions—Assumptions 5–8, compliance neutrality* (2)*, semiconvexity in the sanction* (T6)*, and the forward-looking relief condition* (RL)*—from every initial state: the political state converges to zero in cases* 00 *and* 01*; to a positive fixed point x^{10}_{∞} of the myopic sanction rule $x \mapsto \sigma^{\mathrm{M}}(x, 0)$ in case* 10*; and to a steady state strictly above x^{10}_{∞} in case* 11*.*

The target's patience determines whether relief or non-compliance prevails; the sender's patience shifts the price of relief, or the level at which entrenchment settles. Here x^{10}_{∞} is the long-run state when only the target looks ahead: the state at which the myopic sanction against full non-compliance equals the state. A myopic target accommodates and the state converges to zero, whether the sender is myopic or forward-looking. A forward-looking target maintains non-compliance even against a myopic sender, and sanctions then remain positive and the state converges to a positive level from every starting point. The sender's patience operates on top of this choice. Against a myopic target, a forward-looking sender demands more compliance for relief ($a^{01}(x) > a^{00}(x)$), potentially full compliance, and the myopic target still accepts it. Against a non-compliant target, from a common starting point sanctions in cases 11 and 10 differ by at least the premium bound of Section 3.2 at every date, the forward-looking state path lies above the target-only path, and every steady state of case 11 exceeds x^{10}_{∞} by a strictly positive closed-form margin. The move from the benchmark to the forward-looking equilibrium therefore reflects the target's switch from relief to non-compliance, followed by the sender's foresight premium layered on top of the myopic sanction level. Relief and non-compliance are constructed by different routes. Under non-compliance, the sender's value against full non-compliance is derived first; along the relief path, compliance neutrality pins down the sender's value before the target's policy is known, which is why only relief-related results invoke this condition.

The welfare reversal result of Proposition 9 extends to all four cases. Under the four-case conditions—with a non-decreasing response at non-compliance replacing the increasing-differences condition—two interest-group representations rank the four constructed equilibria $11 \succ 10 \succ 00 \sim$

01 and $00 \sim 01 \succ 10 \succ 11$ ($\succ$: ranked strictly higher; $\sim$: ranked equal) at every starting point and every common planner discount factor. The two cases with zero sanctions follow the same state path, and under compliance neutrality compliance changes no group's total welfare, so they yield equivalent welfare. On the quadratic–linear family, holding the target side and the commercial lobby's exposure fixed and restricting the pro-sanction group's weight to an interval bounds the welfare gap between cases 10 and 11 to an attained closed interval, which yields a strict ranking exactly when it excludes zero.

5.2 Dynamics of the political state

Our analysis also characterizes the path of the political state, and the four cases differ in the long-run level of this path. Without a forward-looking target, the state falls at least geometrically to zero, whether or not the sender looks forward (Proposition 2 and Theorem 12). With a forward-looking target, it converges instead to a positive level: the fixed point x_∞^{10} of the myopic sanction rule when the sender is myopic, and a higher level when the sender also looks forward, along a path that lies strictly above the myopic path at every positive date (Theorem 3). Under any sanction rule that does not decrease with the state, the state and the sanction move monotonically and converge, so the path exhibits no cycles (Proposition E.2 of the Online Appendix).

A sanction maintained for a sufficient duration can shift the state path from the low basin to the high basin. Suppose an equilibrium satisfies the conditions of Proposition 5 for a high basin, and satisfies the corresponding low-basin conditions for a disjoint low basin below it (Section 3.3). Starting from within the low basin, the state remains there and realized sanctions never rise. An exogenously designated sanction changes this trajectory. When full sanctions raise every state below the high basin, a sanction held at maximum intensity for long enough moves the state from any starting point into the high basin, and equilibrium play then keeps it there with sanctions that never decline. A path that would otherwise remain in the low basin stays in the high basin instead (Figure 1). The reverse also holds: a sufficiently long suspension of sanctions moves the state down into the low basin. For an example economy in the Online Appendix, one equilibrium exhibits both basins, and both shocks have explicit durations that suffice from every starting point (Proposition K.1).

Sanctions that begin as bluffs can become self-sustaining under equilibrium play. Under incomplete information, imposing sanctions can serve as a bluff or as a way to call an opponent's bluff. In Spaniel and Smith (2015, pp. 735–736), weaker target leaders have "incentives to bluff strength" by continuing a crisis, and foreign powers respond "by sometimes conceding and sometimes calling potential bluffs by implementing sanctions"; Brekhov (2022, p. 77) analyzes target incentives to bluff under sanctions. In those models, information revealed during the crisis determines subsequent outcomes. Here, sanctions also shift the political state. A sanction imposed as a bluff—or to call an opponent's bluff—and maintained for a sufficient duration acts as a designated sanction in the sense defined above: it moves the state into the high basin, where continued sanctioning is prescribed by equilibrium play. The bluff then becomes self-fulfilling, and sanctions persist without any beliefs to revise.

5.3 Real-world sanctions relief

A natural application of the model is to proposals for sanctions relief, which raise three related but distinct questions; the model's incentive comparisons speak to the first.[18] Would policymakers

[18]The participation result is proved in Appendix C.5 of the Online Appendix; the economy at which a temporary suspension of sanctions benefits a non-compliant target is recorded in the formal supplement.

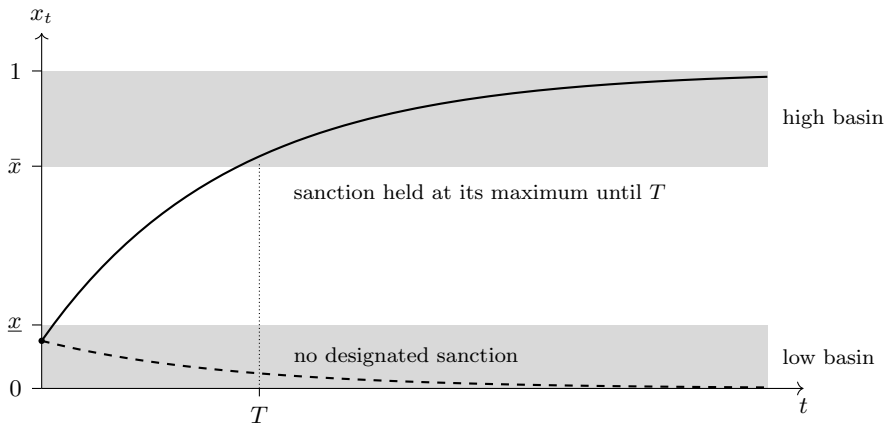


Figure 1: The path of the political state from the same start x_0 in the low basin $[0, \underline{x}]$, under the partial-adjustment law. Dashed: without a designated sanction the state stays in the low basin, where realized sanctions do not rise. Solid: a sanction held at its maximum until date T carries the state into the high basin $[\bar{x}, 1]$, after which equilibrium play keeps it there and the sanction never falls (Proposition 5).

prefer the arrangement and expect it to be honored, and would nominal relief restore the economic activity it is intended to enable? On the first question, consider a temporary suspension of sanctions followed by any equilibrium where the target maintains full non-compliance, such as the constructed equilibrium. Under the entrenchment condition and a sender response at full non-compliance that does not decrease with the state, the change in incumbent payoffs does not support any date-zero side payment between governments that would make the shock mutually acceptable. That neither incumbent gains from such a shock follows jointly from three conditions: the sender's positive marginal return to sanctioning a non-compliant target, the property that compliance does not increase the sender's payoff at zero sanctions, and the entrenchment condition. For an economy failing the entrenchment condition, a temporary suspension of sanctions benefits a target that still does not comply.

The second question—whether relief will be honored—depends on commitment technologies. Lock-in depends on the initial state, and within a high basin resumed equilibrium play generates no endogenous exit under the set and level conditions of Proposition 5. Expiry dates, review clauses, and other commitment technologies operate by altering the strategic environment. Relief poses the hardest test for such mechanisms. Relief that is reversible at no direct cost is the type that Powell (2012, p. 627) identifies as least credible. In recent bargaining models of war, the reliability of an agreement remains a parameter (Ruggiero, 2026). A full model of relief institutions would need to specify, as separate primitives, who can commit, who holds veto power, how private actors respond, and how these choices alter the continuation game.

Commitment problems are one explanation for persistent non-relief among several; the model isolates a distinct mechanism. In this mechanism, continued sanctions can be optimal because of the state that sanctions have built, provided that the sender values that target-side state and the entrenchment condition holds. Sanctions can be optimal even if they are re-optimized every period and a lower sanction level is feasible. Legislative constraints (Halcoussis et al., 2021), administrative persistence, and reputational incentives offer alternative explanations. Political or reputational costs can themselves be incorporated into an optimizing decision framework.

Comparisons across constructed equilibria distinguish two observable responses to a single sanc-

tions regime: greater compliance and deeper entrenchment.[19] Greater compliance arises when the target's forward-looking incentive is weak and a forward-looking sender raises the cost of relief; deeper entrenchment arises when that incentive is strong and a forward-looking sender raises the steady-state level.

North Korea, a context in which this feedback mechanism may be salient, helps motivate the analysis. Haggard and Noland (2017) describe an authoritarian target able to pass sanction costs onto its citizens and a leadership reluctant to weaken its control over foreign economic activity; in the model's terms, these correspond to low sanction pain and a high valuation of the state—the two sides of the entrenchment condition. A United Nations Panel of Experts report documents tighter economic, social, and political controls and more centralized economic management under sanctions (United Nations Security Council, 2024). These same sources document evasion, enforcement, and coordination challenges, which can coexist with the political-state feedback mechanism.

Iran and Russia illustrate all three questions raised above. According to Bajoghli et al. (2024, pp. 115–116), the prospect of earlier sanctions being lifted—not the threat of additional sanctions—brought Iran to the 2015 agreement. Yet the rollback was gradual and required political support at multiple levels of the United States government. Congress, which is often involved in delisting decisions, could impose additional hurdles or reimpose the same sanctions under another pretext, and sanctions remaining on Iran's financial and banking system made the lifting of others immaterial (pp. 121–123). Accepting relief, expecting it to be honored, and being able to benefit from it are therefore distinct margins. In Russia, businesses that have thrived under reduced competition fear the return of Western competitors, while import- and export-dependent firms could gain from partial sanctions relief (Snegovaya et al., 2025, pp. 31–32). The report's scenarios are projections from 2025 (p. 26). This heterogeneity is what the political state variable summarizes.

5.4 Scope and limitations

We have deliberately isolated one public-choice mechanism behind persistent sanctions: sanctions strengthen the target-side constituency that supports their continuation, and both governments value that constituency. Within the class, compliance affects the sender by reducing its incentive to sanction. Given a coercive instrument and a compliance demand, the theorems characterize how equilibrium play evolves from the political state that sanctions—once imposed—have altered. Effectiveness, by contrast, is measured against a sender's objectives, which the U.S. Government Accountability Office identifies for U.S. measures on Russia as economic pressure, inhibition of the war effort, and accountability (U.S. Government Accountability Office, 2025, p. 14). The mathematical structure of the class can be adapted to other coercive systems. Doing so requires mapping their actions, constituency effects, and payoff signs into the class structure, including analogues of each government's stake in the state and of the entrenchment condition.

Existence is established by construction, and each construction pins down its equilibrium only within the sets of value functions it employs. Where the results construct a value function, it is the unique fixed point of its Bellman operator defined over an explicitly stated *invariant set of value functions*: a closed set with stated bounds on the level and slope of each member (and on curvature for the sender), which the operator maps into itself. Section 3 constructs equilibria over the full class of economies. A general existence argument—based on joint contraction of both Bellman equations when the coupling between them is sufficiently weak—follows in the spirit of Stokey et al. (1989) and of broader research on the existence and computation of dynamic feedback Stackelberg equilibria (He et al., 2025; Li et al., 2024).

[19]The model's predictions are collected in Appendix H.9 of the Online Appendix, and the case sources' discussion of evasion and enforcement in Appendix H.7.

Two comparisons delineate the scope of the equilibrium concept: with the stationary strong Stackelberg equilibrium, and with history-dependent strategies. First, target optimality in $\mathcal{E}$ is pointwise: at each state, the target maximizes the right-hand side of (5), taking continuation values as given. In a stationary strong Stackelberg equilibrium, by contrast, the leader selects an entire policy function and anticipates that the follower re-optimizes after every policy deviation; the two concepts need not coincide (Bucarey López et al., 2023; Kudo and Akimoto, 2026). The equilibrium concept is time-consistent because both governments re-optimize at every state. Second, against any equilibrium in $\mathcal{E}$, no pure strategy conditioned on the entire history outperforms the equilibrium's own play, from any initial state and—for the sender—after any observed compliance level on or off the equilibrium path.

Within each equilibrium, the target's compliance rule, the sender's sanction rule, both value functions, both feedback policies, and the political-state trajectory are solved jointly. In the interest-group polity, the pressure levels of competing groups also constitute a Nash equilibrium of the political contest. Two results extend beyond the constructed equilibria under the strong entrenchment condition: the target's minmax payoff bound and the uniqueness of full non-compliance among non-decreasing sanction rules (Section 3.4).

The results operate at three levels of generality. First, some apply to every equilibrium satisfying stated conditions: lock-in (Proposition 5) applies to every equilibrium meeting its basin conditions and, for entry, its transition conditions; the fixed-target-policy comparison (Theorem 4) applies to every non-decreasing sender value at every state, and thus to every equilibrium containing one; the premium bound of Section 3.2 applies to every equilibrium with a non-increasing target policy, for economies in the class with an upper bound on the substitution rate at which larger concessions are never worth more to the sender at zero sanctions; and the minmax bound applies to every equilibrium in $\mathcal{E}$. Second, the myopic-benchmark comparison (Theorem 3), the four-case characterization (Theorem 12), and the threshold result (Proposition 6) derive equilibrium properties and dynamics from inequalities across primitives over the class, and greater target patience preserves the full-non-compliance equilibrium that exists at a lower discount factor (Proposition 7). The welfare results of Section 4 hold across the class, as do the rescaling property and the planner optimality condition for any representation and any bounded welfare flow. Third, some results are established via example economies, which demonstrate that hypotheses can be satisfied or bounds attained: the welfare loss in one interest-group polity (Section 4.3), the attainment of the premium bound (Section 3.2), and economies satisfying the lock-in conditions (Appendix D.10 of the Online Appendix).

Two broad kinds of extensions build on these results: empirical measurement of the political state and its associated payoff terms, and richer theoretical models of the state and of relief institutions. Disciplining estimates of the political-state transition and the payoff terms governing basin entry would require empirical measures that separate the political state from aggregate income, and distinguish within-target adjustment from selection across surviving episodes. Theoretical extensions could incorporate multidimensional political states, stochastic shock duration, forward-looking investment by groups in their own future power (Acemoglu and Robinson, 2008), or explicit relief and commitment technologies. The last of these could alter entry into or create exit from the high basin, and Attia and Grauvogel (2023) document exploratory empirical associations between such provisions and sanction termination. A fully optimal relief policy would require explicit institutional instruments and commitment constraints. A symmetric framework in which both governments directly move the state—of the kind Powell (2013) calls for—would embed compliance directly into the transition law and require rederiving the argument that compliance reduces the sender's incentive to sanction.

6 Conclusion

Economic sanctions can persist because they strengthen the domestic political interests that support their continuation. Without foresight they unravel: in the myopic benchmark, the target accommodates and the political state erodes over time. In the equilibria constructed over the class of economies, when both governments value the political state and are forward-looking, the target maintains non-compliance, sanctions persist indefinitely, and the political state converges to a positive steady-state level. The target's patience determines whether relief or non-compliance prevails; whether the sender is forward-looking shapes the compliance threshold for relief or the intensity of sanctions under non-compliance. These comparisons hold all non-discount primitives fixed. For a fixed target policy, a non-decreasing sender continuation value weakly increases the largest of the optimal sanctions relative to the myopic best response. Under interiority, the constructed equilibria exhibit a strictly positive foresight premium relative to the myopic best response, with a closed-form lower bound.

Beyond the constructed equilibria, patience—like all other target parameters—shapes whether full non-compliance is supportable as an equilibrium. When the sender's response and the target's value under full non-compliance do not decrease with the state, greater target patience preserves the full-non-compliance equilibrium. Under the strong entrenchment condition, the target guarantees itself a minmax payoff against every pure sanction strategy, bounding its payoff in every equilibrium from below; and across the class, every equilibrium with a non-decreasing sanction rule exhibits full non-compliance. In the special case of Proposition 6, a full-non-compliance equilibrium exists if and only if the target's discount factor is at least a threshold value.

Persistent sanctions can be socially costly, and the game alone does not determine their welfare ranking. Across the class, two interest-group representations of the same government payoffs rank the constructed equilibria in opposite orders, and under compliance neutrality one makes persistent play socially optimal. For an example economy with one explicit interest-group polity, equilibrium sanctions remain at their maximum level and generate welfare losses.

The political state can also lock in sanctions. Where an equilibrium and a high-state interval satisfy the conditions of Proposition 5, a state entering the interval remains there under equilibrium play and sanctions never decline. When full sanctions raise every state below the basin, a sufficiently long temporary shock can produce a durable outcome. Persistence can also arise without hysteresis. Where the explicit Lipschitz constant derived for the constructed equilibrium's law of motion is strictly below one, a temporary shock has no permanent asymptotic effect.

Persistent sanctions therefore need not reflect failed bargaining or misperception. Within the model, they can be the optimal choice of governments that value the political state that sanctions themselves have built. With complete information, the levers that end sanctions operate through the political state and government incentives, not through beliefs (Appendix O.4 of the Online Appendix). Whether this channel operates in a given episode is an empirical question: measurements that separate the political state from aggregate income would discipline estimates of its transition and of the payoff terms governing basin entry (Section 5.4). Optimizing governments can thus perpetuate and lock in sanctions through the political state that sanctions generate; patience and other economic features determine whether this occurs.

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CRediT author statement. Xiaofeng Gong: Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, and Writing – review and editing. Yuechuan Zhao: Conceptualization, and Writing – review and editing. Chunping Zhong: Conceptualization, and Writing – review and editing.

Data availability. No empirical data were used. The complete Lean 4 formalization that machine-checks the core results of this paper, as mapped in the formal supplement, is archived as the tagged release `v1.30-submission` of the corresponding author's repository and accompanies this submission as supplementary material: a source archive built from that release by `scripts/Build-CoreBundle.ps1 -Submission`, which excludes the third-party reference libraries and the internal staging documents. The release is public at `https://github.com/qidaolake/political-entrenchment-sanctions-lean`, with the manuscript, the Online Appendix and the formal supplement attached to it; a deposit with a DOI will accompany publication.

Declaration of generative AI in the writing and formalization process. During the preparation of this work the authors used Claude (Anthropic) and Codex (OpenAI) to assist the Lean 4 proof development and the drafting and editing of the manuscript. Each core result is independently verified by the Lean 4 kernel, passing a `#print axioms` check that depends only on `propext`, `Classical.choice`, and `Quot.sound`; no core result relies on the assistants' unverified output. The authors reviewed and edited all content and take full responsibility for it.

Formalization methods. The Lean 4 development was written with assistant-generated suggestions for proof steps and lemma search. The selection of the statements to formalize, the economic interpretation of every formal statement, and the correspondence map between the manuscript and the declarations were made and reviewed by the authors, and the Lean 4 kernel independently checks every proof term in the archived release. A kernel check certifies a formal statement; it does not certify that an informal economic sentence translates that statement faithfully, which is why the correspondence is recorded as a separately reviewed map in the formal supplement.

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